

Chapter 12 Homework3

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Date Assigned: **Nov 25, 2025**

Due Date: **Next class**

Use A4 sheet to answer your questions. Write the answers by hand and submit before next class.

Answer all the questions clearly for a full grade: Each question is worth 20 Pts.

1. Draw time-diagram for 4 bit ripple binary counter.
2. An AB latch operates as follows: If $A = 0$ and $B = 0$, the latch state is $Q = 0$; if either $A = 1$ or $B = 1$ (But not both), the latch output does not change; and when both $A = 1$ and $B = 1$, the latch state is $Q = 1$.

- a) Construct the state table and derive the characteristic equation for this AB latch.

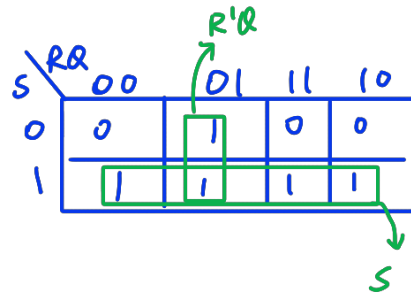
Present state	Next state Q^+			
Q	AB	AB	AB	AB
	00	01	11	10
0	0	0	1	0
1	0	1	1	1

$Q^+ = ?$

- b) Derive a circuit for the AB latch that has **four two-input NAND** gates and two inverters.

3. A set dominant flip-flop is similar to the reset-dominant flip-flop of problem 11.6 except that the input combinations $S=R=1$ sets the flip-flop. Repeat Problem 11.6 for a set-dominant flip-flop. Based on the truth table for the above flip-flop and write the equation for the set dominant flip-flop.

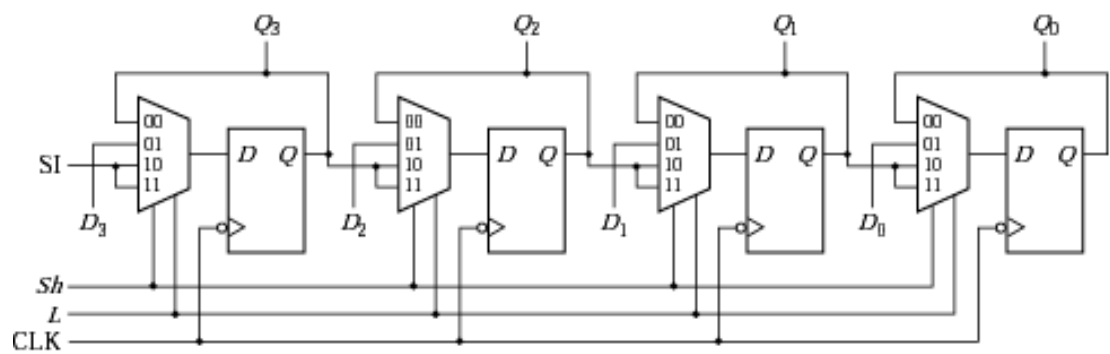
S	R	Q	Q^+
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1



$$Q^+ = S + R'Q$$

$Q^+ = ?$

4. Draw truth table to show shift register operation

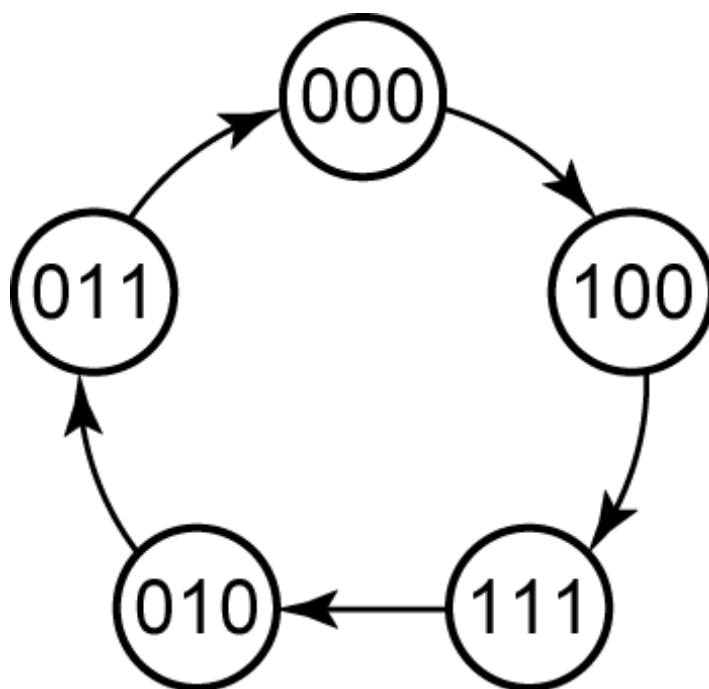


(b) Implementation using flip-flops and MUXes

Inputs		Next state				Action
Shift	Load	Q_3^+	Q_2^+	Q_1^+	Q_0^+	

5. For the same sequence of 0,4,7,2,3,0 repeat

✓ Let us use an S-R flip-flop and derive the counter.



Present			Next State			SR FF inputs		
A_2	A_1	A_0	A_2^+	A_1^+	A_0^+	$S_2 R_2$	$S_1 R_1$	$S_0 R_0$
0	0	0	1	0	0	1 0	0 X	0 X
4	1	0	1	1	1	X 0	1 0	1 0
7	1	1	0	1	0	0 1	X 0	0 1
2	0	1	0	1	1	0 X	X 0	1 0
3	0	1	0	0	0	0 X	0 1	0 1

S	R	Q	Q ⁺
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	X
1	1	1	X

S_2 :

$A_2 \backslash A_1 A_0$	00	01	11	10
0	1	0	X	0
1	X	X	0	X

$S_2 = A_1'$

S_1 :

$A_2 \backslash A_1 A_0$	00	01	11	10
0	0	X	0	X
1	1	X	X	X

$S_1 = A_2$

S_0 :

$A_2 \backslash A_1 A_0$	00	01	11	10
0	0	X	0	1
1	1	X	0	X

$S_0 = A_2 A_1' + A_1 A_0'$

R_2 :

$A_2 \backslash A_1 A_0$	00	01	11	10
0	0	X	X	X
1	0	X	1	X

$R_2 = A_1$

R_1 :

$A_2 \backslash A_1 A_0$	00	01	11	10
0	X	X	1	0
1	0	X	0	X

$R_1 = A_2' A_0$

R_0 :

$A_2 \backslash A_1 A_0$	00	01	11	10
0	X	X	1	0
1	0	X	1	X

$R_0 = A_0$

counter

counter:

